

Abstract

The proposed framework focus on the automated pruning approach of LLMs using combinatorial optimization (pruning is modeled as a combinatorial optimization problem) (this problem is tend to be solved using **Particle Swarm Optimization (PSO)** and **Wheal Optimization Algorithm (WOA)** . (minimize the model size while maintaining the model accuracy)

Pruned model : Llama-3.1-70B

As a result to pruning :

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PSO ( 13.44% size reduction | 19.25% of accuracy degradation )  
WOA ( 12.07% size reduction | 22.81% of accuracy degradation )
```

post-process :

```
fine-tine the pruned model using spatialized algorithms to recover some  
accuraccy degradation
```

- **As a result :**

PSO (12.72% of accuracy degradation "from the original model")

WOA (14.83% of accuracy degradation "from the original model")

1. Introduction

- the utilization of LLMs on the large scale requires a massive hardware deployments (larger VRAMs for GPUs or TPUs , along with considerable RAM and storage capacity to handle the massive amount of weights and computations (ex : Llama-[7B-405B] requires GPUs with VRAMs - [140GB -> 70B variant - FP precision] /[35GB with 4-bit quantization]))

several approaches exist to minimize the size of the model :

1. knowledge distillation : works by transferring information from the teacher model into a smaller student model , however it requires massive computational resources (you are performing training)
2. Quantization : drastically compresses the model size , but it degrade the accuracy
3. Traditional pruning : redundant parameters can be removed to effectively reduce the model size and computational resource usages , however choosing which parameters to prune is the challenge .

- traditionally selecting the optimal subset of parameters to prune is practically infeasible due to the vast number of possibilities (exponential search space of parameter subsets and the interdependencies between parameters) , therefore evolutionary algorithms such as WOA , PSO ,ant colony optimization (ACO), genetic algorithm (GA), and others are commonly applied as heuristics in selecting the optimal candidates to prune .
- traditional pruning techniques often uses user-defined heuristics that may not generalize well over different architectures and datasets , in this paper they use evolutionary algorithms to iteratively optimize the pruning strategy which selects and removes the model parameters based on their contribution to the model performance , these are WOA and PSO which proved their ability to provide a balanced tradeoff between the model accuracy and reduced size

2. Related Work

2.1. Large Language Models

Table 1. Comprehensive comparison of leading large language models (LLMs).

Model	Author	Performance	Use Case	Params	Variant
GPT	OpenAI [11]	High accuracy, low perplexity	Creative writing, chatbots, coding	1.8 T	GPT-4o
				N/A	GPT-03-mini
Gemini	Google [12]	Strong reasoning	Creative Applications	>200 B	Gemini 1.5 pro
Claude	Anthropic [13]	Extended thinking, strong safety	Safety-critical applications	N/A	Claude 3.7 Sonnet
				N/A	Claude 3.5 Haiku
				N/A	Claude 3 OPUS
Mistral	Mistral AI [14]	Efficient, Low params	Real-time apps, programming	8 B	Ministral 8 B
				24 B	Mistral Saba
LlamA	Meta [15]	Competitive scores	Research, academia, open source	1 B, 3 B, 11 B, 90 B	Llama 3.2
				70 B, 8 B, 405 B	Llama 3.1

Llama-3.1-70B where used for this study where it balances performance , accessibility, and transparency in AI research (open source) . Its architecture incorporate several state-of-the-art techniques such as multi-head self-attention and layer normalization to achieve high performance

with 70 billion parameters it is considered as the second-largest model in the Llama3.1 series , allowing it to capture a broader range of linguistic nuances and complex patterns .

2.2. Model Compression Techniques

compressing neural network models is crucial strategy for reducing their size and computational complexity , however this need not to be at the expense of the model performance . There are variety of pruning strategies each with its pros and cons (pruning , quantization , knowledge distillation , multiplication reduction , resource efficient architectures , low-rank adaptation (LoRA) , and **network architecture search ???**)

Pruning :

It involves removing redundant weights or neurons from the model , therefore leading to a significant reduction of the memory usage and computational overhead . but this must be done cautiously as it can lead to accuracy degradation .

Quantization :

this involves reducing the numerical precision representation of the model's weights and activation values , example , from 32-bit floating point representation to smaller precision representation 'ex : 8-bit' , this will drastically reduce the storage requirements and speedup the inference time for hardware supporting integer operations like INT-8 cores on GPUs, but also introduce quantization noise which degrade the accuracy (drastically for models with high sensitivity to small weight variations)

knowledge distillation :

involves training a smaller and efficient model called the student model to replicate the behavior of a larger more accurate model called the teacher model . Despite its being reduced size the student model can achieve promising performance close to the teacher model , however this may not work well in case there is a significant performance gap between these models , as the student may struggle to generalize complex patterns learned by the teacher .

Others :

/ ----- to be filled later

over all of these methods the pruning approach is a direct and intuitive way , as it consumes relatively low computational overhead compared to others like Network architecture search (NAS) and knowledge distillation , it does not change much the network architecture and does not require retraining process (carefully prune few parts of the network and finetune the remaining components).its possible to achieve significant model compression with retaining high accuracy .

2.3. Pruning Methods

magnitude-based :

remove the smallest weights assuming they contribute less to the performance

structured:

removes entire neurons , layers , channels which optimizes hardware efficiency but requires retraining

unstructured:

removes individual weights that lead to sparse models which require spatialized hardware

global pruning :

selects the least important weights network-wide , its more effective but computationally expensive

local pruning :

applies threshold per layer

iterative pruning :

focus on iteratively (training cycles) removing parameters(gradually) while maintaining accuracy , but increasing training time

layer pruning :

focus on selectively removing entire layers or blocks of neurons ,this will help remove redundant computations and accelerate execution of the model without complex parameter-level adjustments , however limiting the search space to entire layers rather than parameters will restrict the model ability to adjust its structure for optimal performance .

Table 2. Comparison of pruning methods.

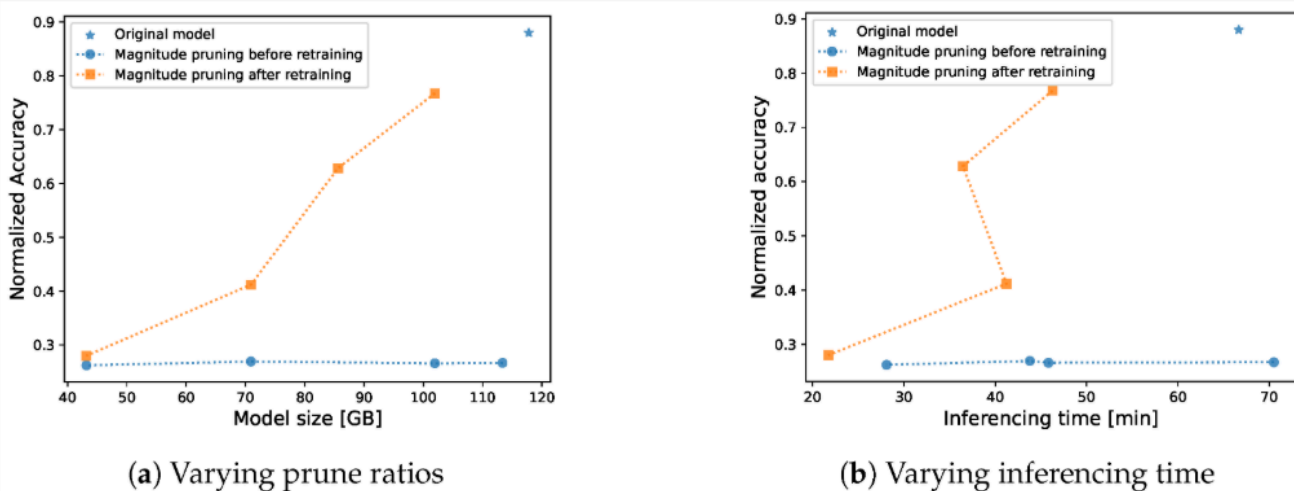
Pruning Method	Advantages	Disadvantages
Magnitude-based Pruning	Removes least important weights	Requires retraining to recover accuracy, may remove the important parameters
Structured Pruning	Optimizes hardware efficiency, improves inference speed	Requires retraining
Unstructured Pruning	Leads to sparse models, can greatly reduce parameter count	Requires specialized hardware for efficient execution
Iterative Pruning	Helps maintain accuracy by gradual pruning	Increases training time due to multiple pruning cycles
Global Pruning	More effective in removing unimportant weights across the model	Computationally expensive and difficult to tune
Local Pruning	Easier to implement with per-layer thresholds	Less optimal than global pruning for overall efficiency
Layer Pruning	Reduces model complexity significantly, improves execution speed	Requires retraining

many studies have proposed this layer pruning (using heuristics)including layer pruning via structured residual (LPSR) . Zhang et al . The pruning process involves estimating the impact of each block using Taylor expansion and batch normalization (BN) parameters, ensuring that only the least important blocks are removed.

two other methods also , the first proposed a method to select the pruned layer through one-shot layer evaluation using proxy classifier (tiny classifier attached at the end of each layer) and imprinting (way to initialize the weights of these classifiers) , then as a result the layer with lower accuracy contribution is pruned . The second method proposed stochastic re-initialization approach (re-initialize a randomly selected layer each time) , given the importance of each layer to the sensitivity in stochastic re-initialization (low accuracy drop) (ie each layer that has low sensitivity to the accuracy is dropped (pruned))

- magnitude layer pruning may have good potential in preserving accuracy ,but it has limited effectiveness without retraining . even though the accuracy is preserved the computational latency change may be marginal , as some of the pruned layer were not computationally expensive in the first place . and some layers are even contributing to the computational efficiency .

Figure 2. Normalized accuracy for varying prune ratios (a) and inferencing time (b).



PSO and WOA were chosen for their ability in solving high-dimensions combinatorial optimization problems ; PSO demonstrated strong global search capability while WOA shows a balanced capability between exploration and exploitation . GA (Genetic algorithm) and SA (simulated annealing) have significant evaluation and search times which make them not best suited in pruning optimization . GA suffers from slow convergence due to the stochastic nature of crossover and mutation operations ; SA also has slow cooling schedules which make the convergence slower .

PSO is inspired by the collective operation of animals , like swarm of birds , where each individual in a swarm adjusts its position based on its own experience and experience of its neighbors , this enables the swarm as a whole to converge toward the global optimum , its strength lies in its ability to balance between exploitation 'refining known solution ' and exploration 'searching the search space ' .

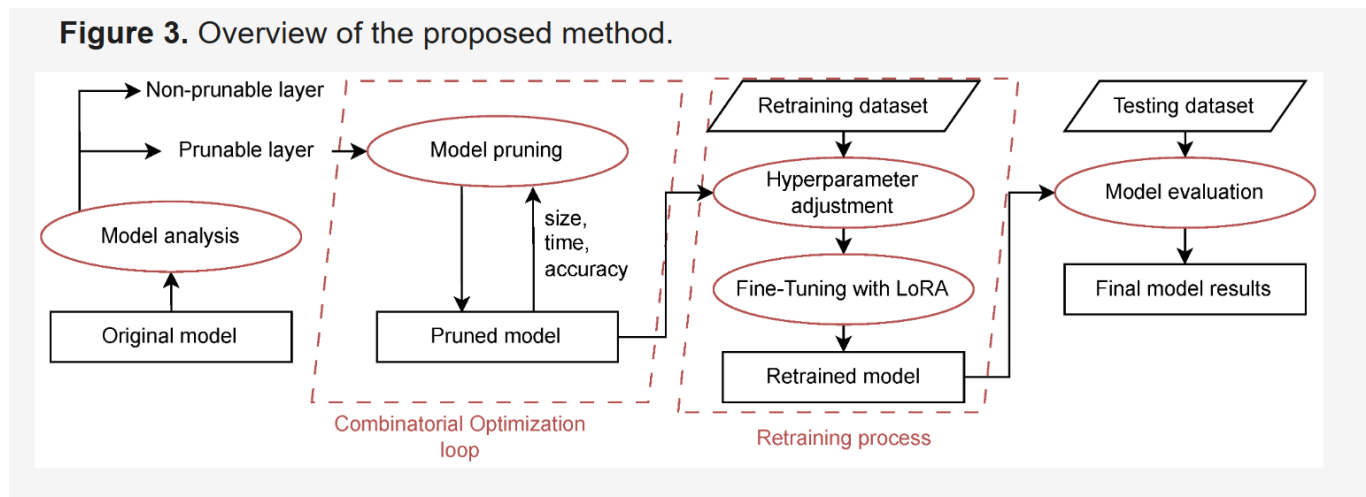
WOA is inspired from the behavior of the humpback whales who create a spiral pattern in the water to trap fish , WOA mimics this spiral movement by implementing a series of mathematical operation . its ability to perform random exploration effectively allows it to escape premature convergence and stuck in local optima like other algorithms , as well as maintaining a robust search in complex search spaces . (its ability to blend exploration and exploitation to prevent premature convergence make it a strong candidate for high-dimensional or multi-modal problems)

PSO and WOA offers a promising approach for automated layer pruning , they can search high dimensional and large spaces to prune identifying the most suitable pruning configuration with the least impact on the model performance . Beside it improves the model efficiency it also saves time and computational resources required for manual pruning .

3. Materials and Methods

1. analyze the model and select the layers to prune
2. prune the selected layers using the combinatorial optimization iterative process
3. retain the pruned model using dataset different from the one used in the original model training
4. evaluate the pruned and retrained model

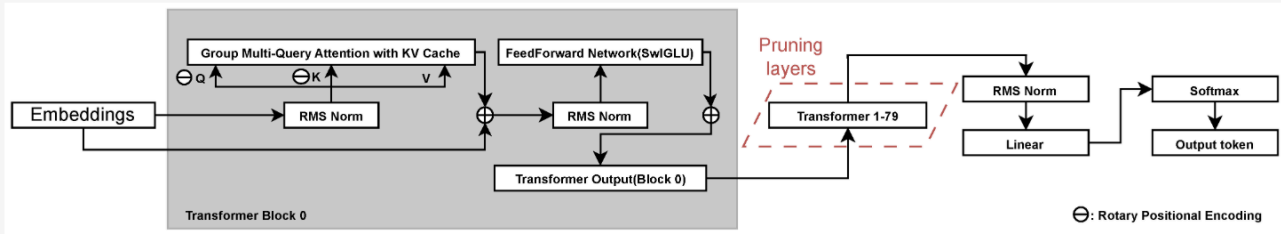
Figure 3. Overview of the proposed method.



3.1 Model Analysis

Llama-3.1-70B is a large language model developed by Meta that is based on Auto-Regressive architecture . its architecture consists of the main components . The embedding layer (responsible for the converting of the input tokens into dense vector representation that captures semantic information) , follower by transformer blocks (80 core transformer blocks standing out as the main computational units) , and finally by a final normalization layer follower by a linear layer .

Figure 4. Llama-3.1-70B model structure.



the transformer blocks are stacked sequentially each consisting of multi-head self-attention mechanisms and position-wise feed forward neural network . they are responsible for capturing complex dependencies and hierarchical patterns within the input data , after that their output is passed through normalization layer to stabilize the training and finally trough a linear layer to produce the final output . these 80 transformers directly correspond to the layers .

Three main components in Llama's layer :

1. Multi-Head Self-Attention Mechanism : enabling the model to capture a wide range of complex contextual relationships by projecting the inputs to multiple attention subspaces
2. Feed Forward Neural Network : position-wise feed forward layers in each transformer block to boost the model learn more complex relationships
3. Rotary positional Embeddings (RoPE) : enables the model to process long sequences by including relative positional information directly into the attention .

3.1 Model Pruning

as described before the problem of selecting which layers to prune or retain is treated as combinatorial optimization problem.

The Total number of possible pruning configuration is described by the following equation :

$$TotalConfigurations = \sum_{i=0}^k \binom{80}{i}$$

3.3. Combinatorial Optimization Algorithms

The search space is defined as a subset selection problem , where we configure the number of pruned layer as input .

$$S = \{l_1, l_2, \dots, l_k\}, \quad \text{where } l_i \in \{1, 2, \dots, 80\}, \quad k = \text{number of pruned layers.} \quad (2)$$

For example, when pruning 10 layers, the variable S will be

$$S = \{3, 7, 12, 15, 19, 24, 28, 30, 35, 40\} \quad (3)$$

Layers 3, 7, 12, ...40 will be pruned while others are retained.

A fitness function is used to penalizes severe accuracy loss while encouraging model sparsity.

3.4. Model Retraining

the retraining process follows two stages :

1. Hyperparameter Adjustment: to improve stability and convergence , key hyperparameters like learning rate , epochs , batch size are carefully tuned .
2. Fine Turning with LoRA : in addition of retaining with general-domain dataset, instead of full-parameter training , Low rank adaptation method is used to inject trainable parameters in the form of low-rank matrices , to effectively help the model recover its accuracy and prevent overfitting (by focusing on small set of parameters) .

3.5. Model Evaluation

The Framework leverages The Language model evaluation harness , an open-source toolkit that provides a standardized and flexible benchmarking setup for LLMs , compatible with models from the Hugging Face ecosystem . as long the model is supported by the evaluation harness and it supports transformer based architecture the purposed framework generalizes well .

in this study they use the HellaSwag dataset (for common-sense reasoning tasks) . the evaluation process measures (accuracy , size and inference time)

4. Experimental Results

Hardware used :

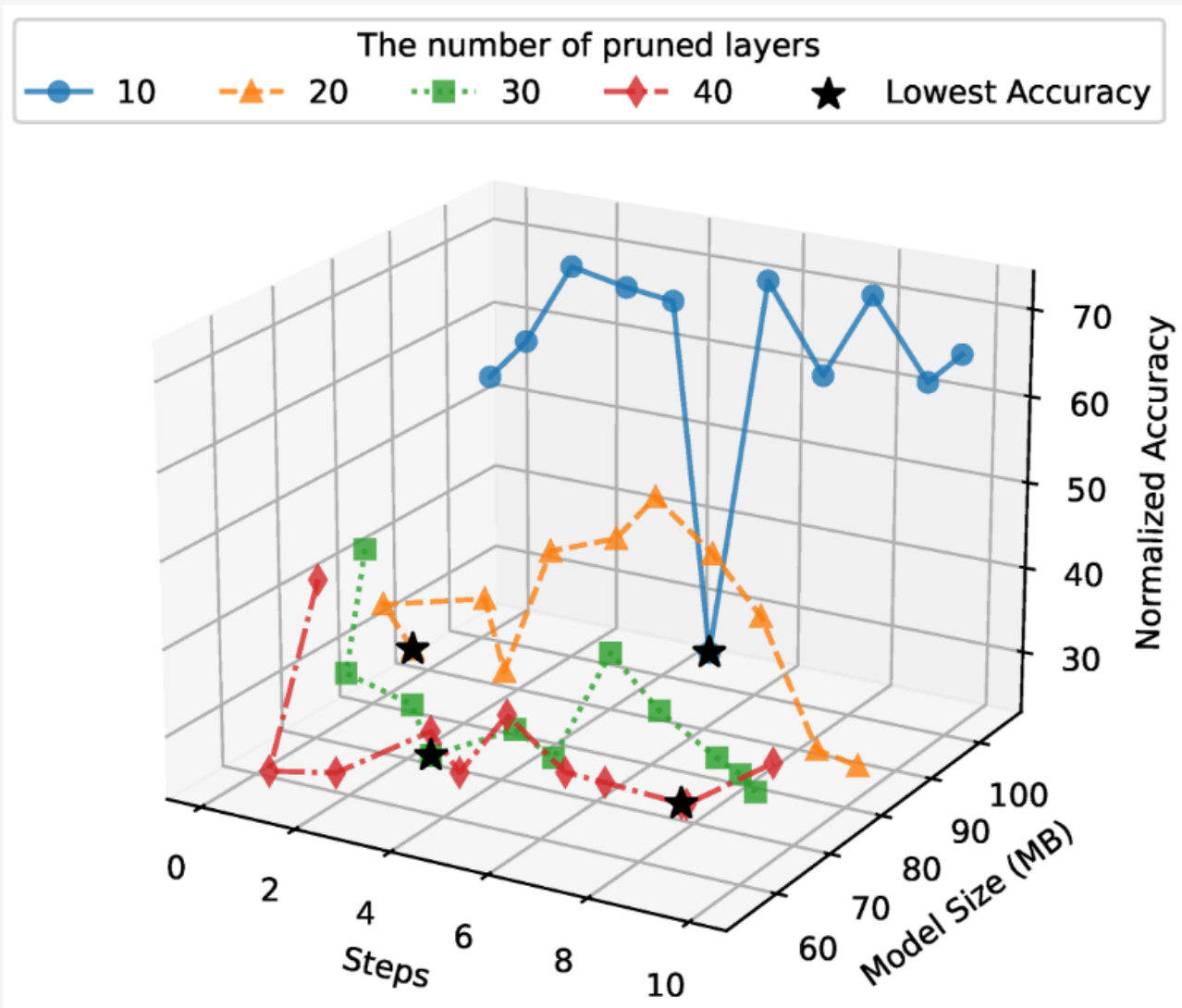
Table 3. Hardware specification.

Hardware	Specification
CPU	Xeon Gold 62.30R 26 core ×2
Main Memory	256 GiB
GPU	NVIDIA A100 ×2
GPU Memory	40 GiB ×2

the optimization process was carried out using **Mealpy** library , and weighted sum to combine multiple objectives for best exploration of the trade-offs between the competing factors (size , accuracy)

4.2. PSO Results

Figure 5. Model's size and normalized accuracy during PSO optimization.



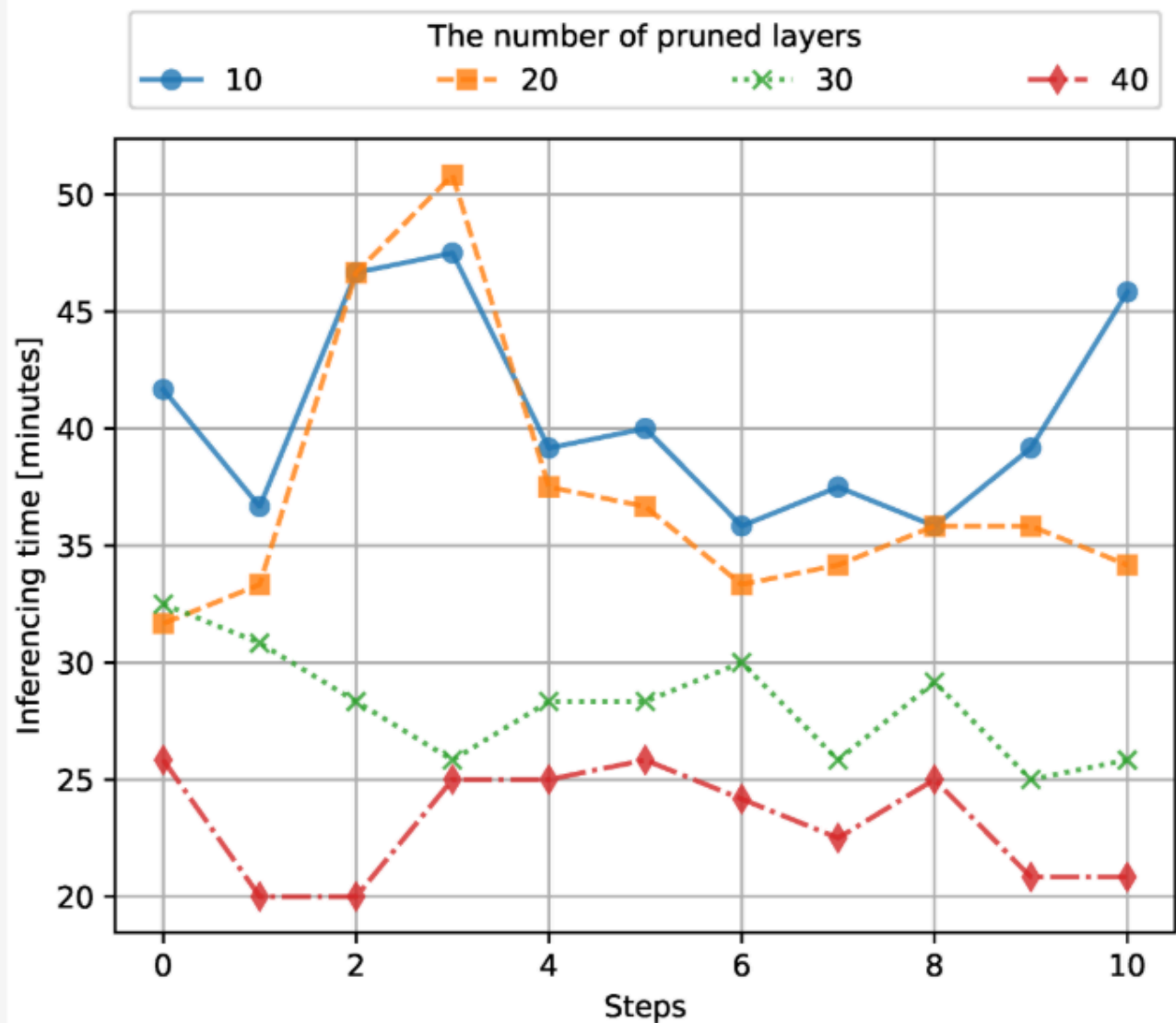
PSO progressively balanced trade-off between maintaining model accuracy and both model size and inference time . normalized accuracy is adjusted using **byte-length normalization** (probability of continuation is divided by number of bytes it occupies) "**consistent scoring** across **different tokenization schemes**).

- **10 pruned layers** : model still maintain most of its accuracy over optimization steps (**occasional drops** during **steps 8,5**)
- **20 pruned layers** : in this case the accuracy exhibits grater fluctuations (peak at step 6) , then significantly degrading afterwards
- **30 pruned layers** : the normalized accuracy becomes more unstable (more fluctuations) , it followed the same pattern as 20-Layers however this implies that as more layers are removed the model struggles to maintain its predictive capability .
- **40 pruned layers** : the model exhibits the lowest and most unstable case of accuracy values .Excessive pruning result in a drastic degradation in the model's ability to generalize effectively .

After investigation , the following observations are made :

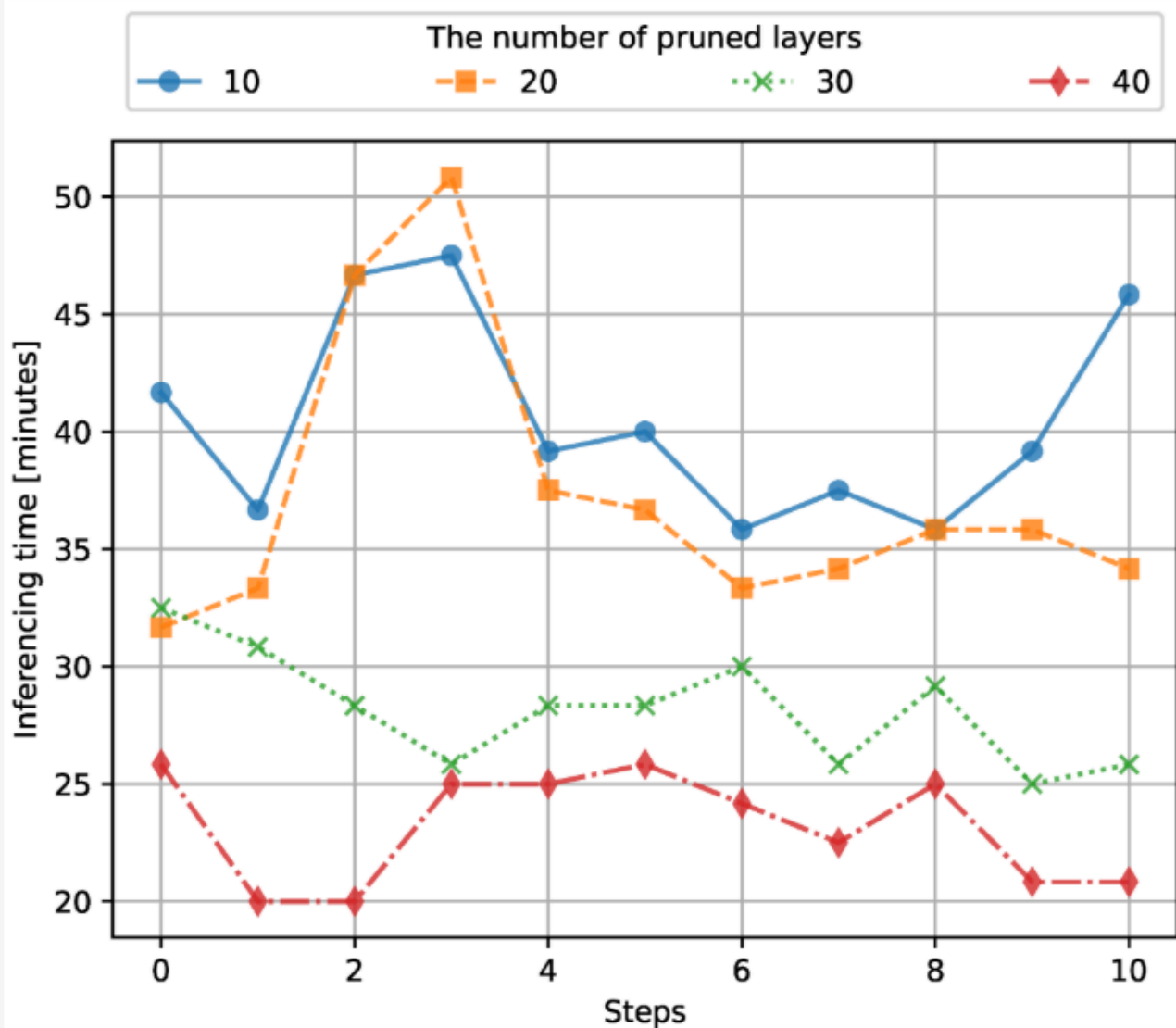
1. **7 Layers** consistently appears in the steps with **lowest accuracy** , and being selected three timers across different steps , which indicated their importance in preserving model accuracy .(eg in case of 10-Layer pruning , 3 of those where removed in the 5th step)
2. **52nd and 37th** where most frequently removes in the steps with **highest accuracy** indicating their important in preserving the model performance (computation overhead) .

Figure 6. Model's inferencing time during PSO optimization.



- the 38th layer where selected for pruning in every configuration that result the smallest model size , which indicate its big contribution to reducing the model size .

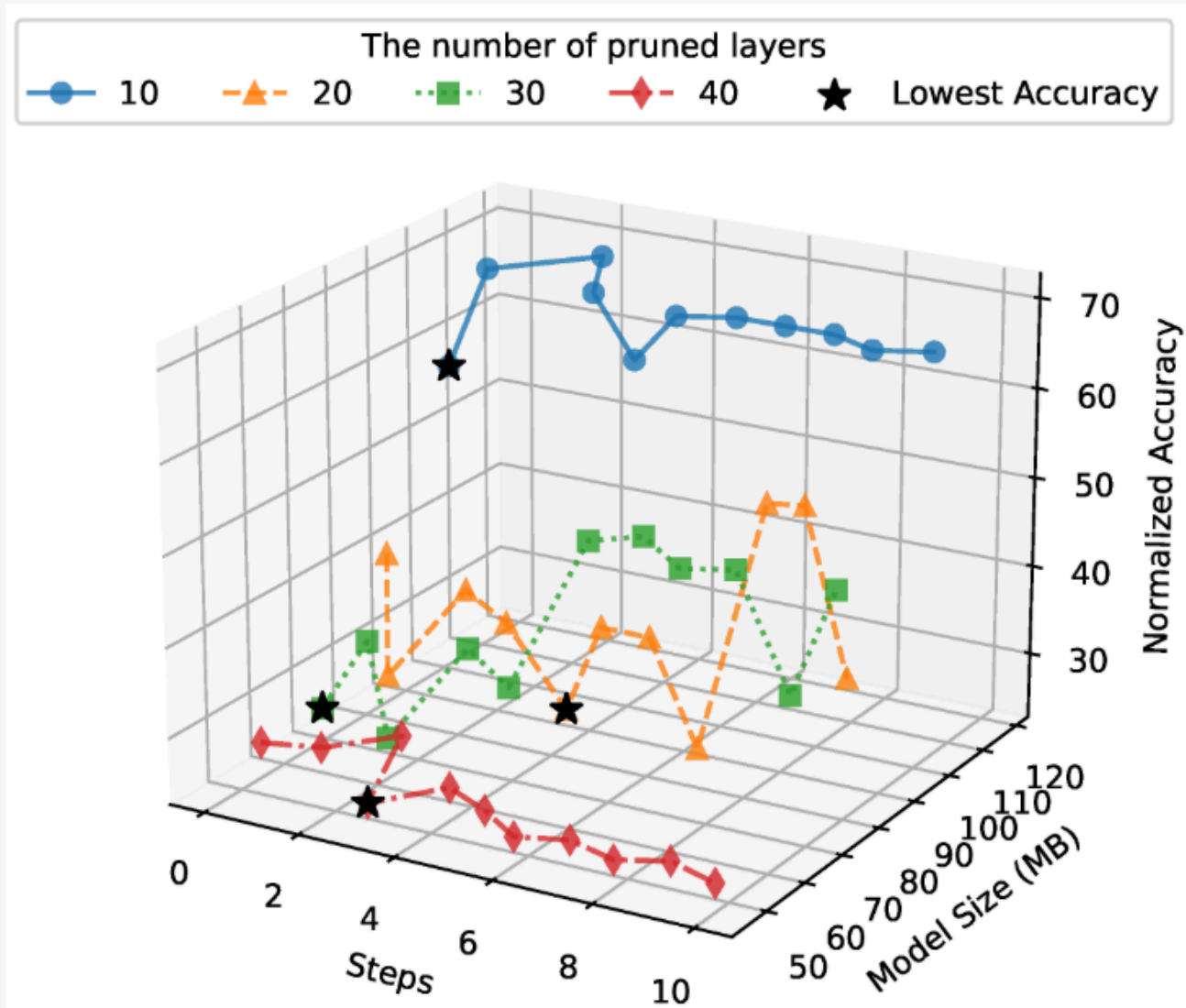
Figure 6. Model's inferencing time during PSO optimization.



4. the 38th and 46th layers were most pruned in the optimization steps associated with higher inference times. the 38th layer reduces the model size but also increases the inference time
5. over the whole optimization process, layers located in the middle part of the model were most selected for pruning '37th, 40th, 45th, and 46th layers -> 20 times' which are considered more redundant, and those located near the input and output were less selected '0th, 75th, 76th, and 78th layers -> 1 time' which indicate their importance in preserving accuracy.

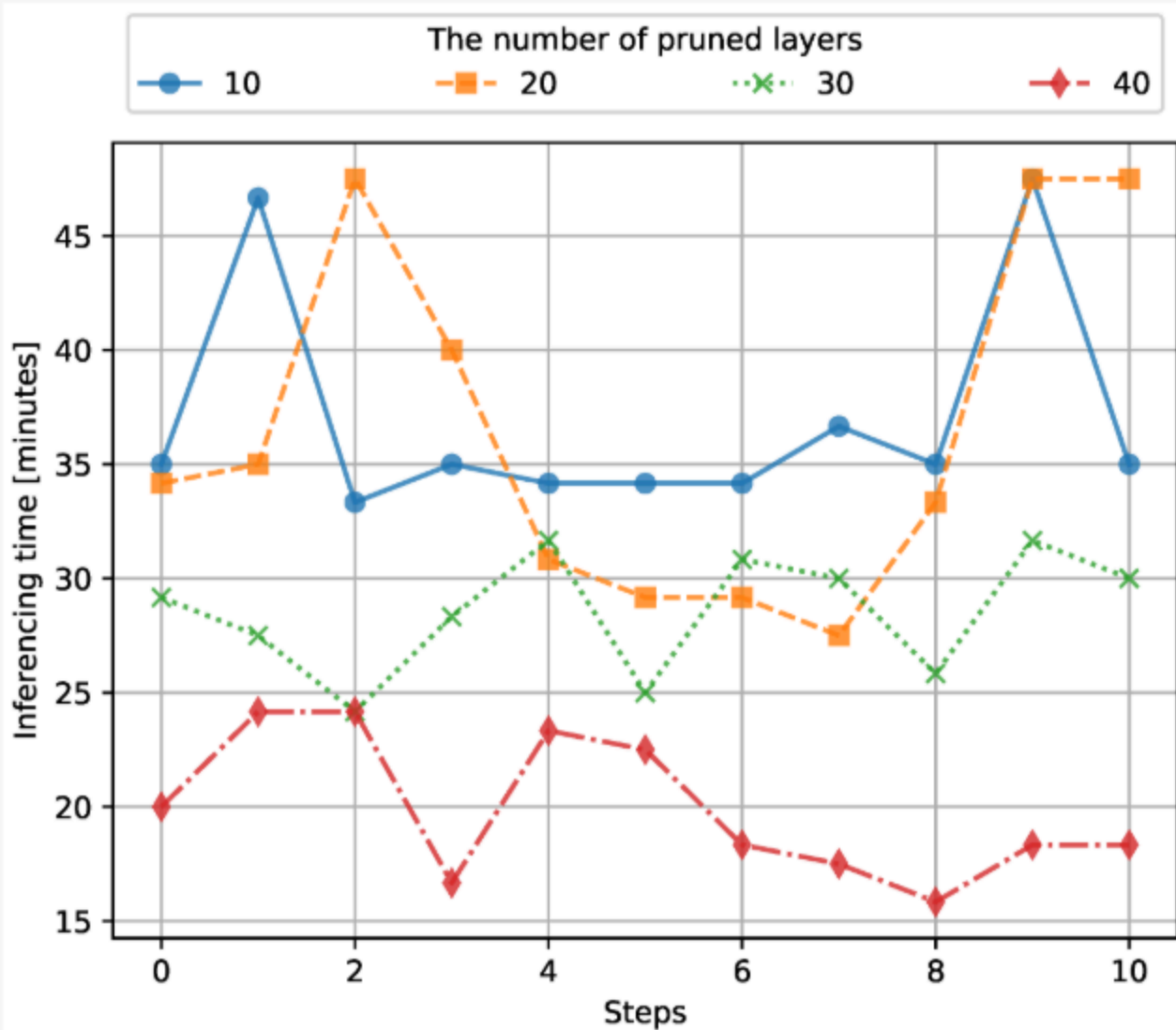
4.3. Results for Pruning with WOA

Figure 8. Model's size and normalized accuracy during WOA optimization.



1. 5 Layers were frequently selected across steps with lowest accuracy ' each appearing 3 times ', 25th layer overlapped with PSO's result
2. the 14th, 16th, 41st, 56th, and 60th where the most frequent pruned in the steps with highest accuracy .
3. The pattern of accuracy stabilization in WOA like PSO , indicates its effectiveness in preserving the model accuracy while pruning .
4. 22nd layer was the most consistently pruned leading to the smallest model size in each pruning step ; 38th layer also had a significant impact on reducing model size, similar to the PSO results.
5. from PSO and WOA results => 38th layer stands out as particularly influential in size reduction

Figure 9. Model's inferencing time during WOA optimization.



6. 24th and 33rd where frequently selected in steps with longer inference time .
7. 22nd, 32nd, and 44th layers were predominantly selected in the steps with shorter inference times.
8. The relationship between the layer selection and inference time is complex (some layers influence the inference time in unexpected way)
9. results from both PSO and WOA shows that 10-Layer pruning maintains the highest and most stable accuracy , 40-layer leads to lowest accuracy across all steps .
10. The accuracy fluctuations are more pronounces in PSO (20- and 30-layer pruning configurations) while WOA demonstrate smoother variations in accuracy indicating its ability to maintains more consistent performance .
 1. WOA shows more fluctuations in size (layer-20 ; layer-30) compared to PSO ; but smother and stable once in accuracy and inference time compared to PSO . PSO has grater global search capability allowing to escape local optima more effectively , while

WOA's smoother optimization process suggests that it may be best suited for scenarios where stability in accuracy, size, and efficiency is critical.

- the pruning pattern is similar between PSO and WOA with just WOA demonstrating sharper peaks than PSO .

4.4. Comparison Between PSO and WOA

both algorithms tend to focus on middle layers (21–59) likely because these layers contribute less to overall model performance . PSO exhibits evenly distributed range across the middle layers . with peak around 40th layer . WOA shows sharper peaks (particularly around the 25th and 50th layers) , both algorithms prune edge layers (0–20 and 60–80) less frequently, indicating they are more critical to maintaining accuracy .

Table 4. Comparison of pruned model using PSO and WOA after retrain.

Models	Avg Search Time (Min)	Red. Model Size (GB)	Infer. Time (Min)	Normalized acc.
PSO(10)	40.5303	101.9299	50.6833	0.7680
WOA(10)	36.9697	103.5619	49.4834	0.7495
PSO(20)	37.2727	74.1854	32.9	0.6274
WOA(20)	36.5151	74.1854	37.0667	0.2676
PSO(30)	28.1818	51.3369	24.5166	0.4239
WOA(30)	28.5606	51.3369	22.5333	0.4888
PSO(40)	23.1818	20.2383	13.7167	0.2676
WOA(40)	19.9242	20.2383	20.0	0.2637

1. Search Efficiency (Time)

- **Winner: WOA**
- **Finding:** WOA is faster at traversing the search space. For a 40-layer prune, WOA took **19.92 minutes** compared to PSO's **23.18 minutes**.
- **Trade-off:** While WOA is faster, its "high-speed" search can sometimes result in lower accuracy if too many layers (e.g., 40) are removed.

2. Model Size Reduction (Compression)

- **Winner: Tie (Slight edge to PSO)**

- **Finding:** Both algorithms achieve similar compression levels. However, at lower pruning levels (10 layers), **PSO achieved a slightly smaller model size** (101.93 GB) than WOA (103.56 GB).

3. Inference Speed (Execution Time)

- **Winner: Inconsistent**
- **Finding:** There is no clear winner for speed.
 - At 20 layers, PSO was faster.
 - At 30 layers, WOA was faster.
- **Insight:** This proves that inference speed isn't just about how many layers you remove, but **which specific layers** are removed. Some layers are more "computationally heavy" than others.

4. Accuracy & Scaling

- **Key Discovery:** The "Best" algorithm changes based on how much you prune:
 - **For Heavily Pruned/Smaller Models (30 layers removed):** WOA is superior, outperforming PSO's accuracy by **15.31%**.
 - **For Lightly Pruned/Larger Models (10 layers removed):** PSO is superior, outperforming WOA's accuracy by **2.47%**.
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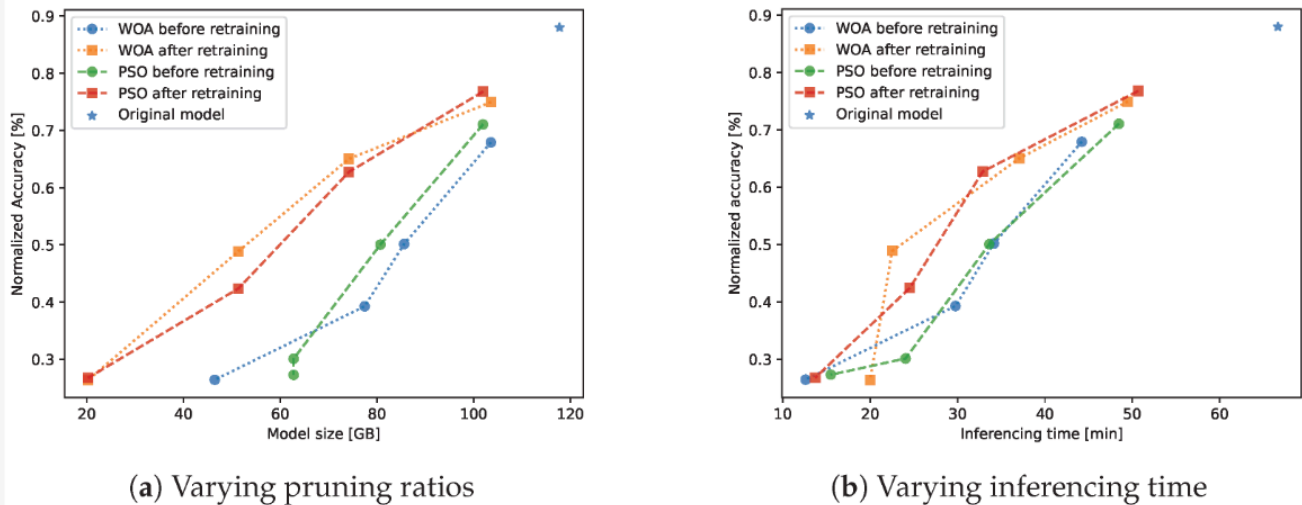
Final Recommendation from this text:

- **Use WOA** if the goal is a **smaller model** (under 75 GB) or if you need to prune a large number of layers (30+).
- **Use PSO** if you are working with **larger, high-parameter models** (where you only prune ~10 layers) and accuracy is the top priority.

4.5. Compared to Original Model

By merging LoRA weights directly into the pruned architecture during retraining, the framework significantly decreases storage and computational demands while successfully maintaining the model's performance characteristics.

Figure 10. Normalized accuracy for varying pruning ratios (a) and inferencing time (b), comparing algorithms with and without retraining.



1. Discussion: The Path to Dynamic AI

The discussion shifts from what the framework did to what it needs next to be used in the real world:

- **From Static to Dynamic:** Current pruning is "static" (done once). The next step is **on-device, dynamic pruning** where the model adjusts its own size while you use it.
- **Resource-Awareness:** Future versions must include "lightweight monitoring modules" that check the device's **battery life, available RAM, and CPU heat** in real-time to make pruning decisions.
- **User-Driven Evolution:** The framework should allow models to "evolve" based on how a specific person uses them—pruning parts of the model that the user rarely interacts with to save power.

2. Conclusion: Final Verdict & Results

The conclusion summarizes the "Winning Strategy" for reducing Llama-3.1-70B:

- **Quantitative Success:**
 - **PSO** was the overall winner for large-scale models, achieving a **13.44% size reduction** with only **12.72% accuracy loss**.
 - **WOA** was slightly behind in size reduction (**12.07%**) and had higher accuracy loss (**14.83%**).
- **Algorithm Selection Rule:**
 - **Use WOA** for "Small-to-Mid" models (under 75 GB).
 - **Use PSO** for "Large" models (70B parameters and up).

- **The "Middle-Layer" Rule:** The most important discovery is that layers **0–20 and 60–80 are critical** for the AI's "brain," while layers **20–50 are redundant** and can be safely removed to save space.
 - **Future Work:** The team plans to combine this pruning with other techniques like **Quantization** and measure **actual power consumption** on physical mobile hardware.
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Key summary

1. Introduction & Methodology

- **Problem:** LLMs like Llama-3.1-70B are too large for edge devices. Pruning is modeled as a **Combinatorial Optimization** problem.
- **Search Space:** With 80 layers, pruning up to layers creates a space of configurations.
- **Framework:**
 1. **Analyze:** Llama architecture (80 blocks of Self-Attention, FFN, and RoPE).
 2. **Prune:** Automated selection via PSO or WOA using a weighted-sum objective.
 3. **Retrain:** Two-stage recovery (Hyperparameter tuning + **LoRA fine-tuning**).
 4. **Evaluate:** Standardized testing via HellaSwag dataset.

2. PSO (Particle Swarm Optimization) Investigations

- **Accuracy Behavior:** Stable at 10 layers, but exhibits **high fluctuations** and instability at 20–30 layers. Drastic degradation occurs at 40 layers.
- **The "Layer 38 Paradox":** Investigation showed Layer 38 is the most influential for **reducing size**, but pruning it actually **increases inference time**.
- **Inference/Size Stability:** Unlike its accuracy, PSO's size reduction and inference trends were relatively **stable and consistent** across steps.
- **Critical Layers:** 7 specific layers were identified as "untouchable"—removing them caused sharp accuracy drops.

3. WOA (Whale Optimization Algorithm) Investigations

- **Accuracy/Inference Behavior:** Demonstrated **smoother and more stable variations** than PSO, making it better for consistent performance.
- **Size Behavior:** Conversely, WOA showed **higher fluctuations in model size** (specifically at the 20 and 30-layer marks) compared to PSO.
- **Key Layer:** Layer 22 was identified as the most pivotal for size reduction in WOA configurations.

- **Overlap:** Investigation found that both algorithms identified **Layer 25** as a critical shared point for maintaining accuracy.

4. Comparative Analysis & The "Middle-Layer Rule"

- **Commonality:** Both algorithms consistently targeted the **Middle Layers (20–50)** as redundant.
- **Edge Sensitivity:** Layers **0–20** and **60–80** were rarely pruned, confirming they are essential for the model's structural integrity and intelligence.
- **Search Efficiency:** **WOA is faster** (19.92 min vs. 23.18 min), but can sacrifice accuracy at high pruning rates.
- **Performance Scaling:**
 - **WOA** is superior for **smaller models (< 75 GB)** or deep pruning (30+ layers).
 - **PSO** is superior for **larger models (70B+)** where precision in layer selection is critical.

Metric	Winner	Insight
Search Speed	WOA	Faster at traversing the search space.
Large Model Accuracy	PSO	Better at preserving complex dependencies in 70B+ models.
Inference Time	Inconsistent	Depends on which specific layers are pruned, not just how many.
Size Reduction	PSO	Achieved slightly better compression at lower pruning levels.

5. Discussion: The Future of Dynamic AI

- The framework currently performs **static pruning**, but the authors propose moving toward **Dynamic, On-Device Pruning**.
- This involves **Real-time Monitoring** of hardware (battery, RAM, heat) to prune the model "on the fly" based on user behavior and device constraints.

6. Conclusion

- **Final Results (70B):** **PSO** achieved a **13.44% size reduction** with a recovered accuracy loss of only **12.72%**.
- **Strategy:** Automated pruning is a viable foundation for model compression. Future work will integrate these algorithms with **Quantization** to further minimize power consumption on physical hardware.

Final Comparison: PSO vs. WOA

Metric	PSO (Particle Swarm)	WOA (Whale Algorithm)	Key Finding
Search Time (40 Layers)	23.18 Min	19.92 Min	WOA is faster at traversing the search space.
Model Size Reduction	13.44%	12.07%	PSO achieved higher compression on the 70B model.
Accuracy Loss (Final)	12.72%	14.83%	PSO is better at preserving accuracy for large-scale models.
Accuracy Stability	High Fluctuations	Smooth & Stable	WOA is more consistent in maintaining accuracy steps.
Size/Inference Stability	Consistent & Stable	Higher Fluctuations	WOA size results varied more at the 20/30 layer marks.
Target Layer Overlap	Middle (20–50)	Middle (20–50)	"Middle Layer Rule": Both agree the center is redundant.
Best Used For...	Large Models (70B+)	Smaller Models (< 75 GB)	Algorithm effectiveness scales with model size.